

Duration: 3hrs

[Max Marks:80]

- N.B. : (1) Question No 1 is Compulsory.  
(2) Attempt any three questions out of the remaining five.  
(3) All questions carry equal marks.  
(4) Assume suitable data, if required and state it clearly.

- 1 Attempt any FOUR [20]
- a List and explain the challenges in Generative Models
  - b Differentiate between Bagging and Boosting ensemble technique
  - c Differentiate between Generative and Discriminative modeling
  - d Explain any two challenges associated with Generative Adversarial Network
  - e Differentiate between Virtual Reality and Augmented Reality
- 2 a Explain self-supervised learning and meta learning with suitable examples [10]
- b Explain Random Forest algorithm with suitable example [10]
- 3 a Write a short note on Hidden Markov Models with suitable example [10]
- b Explain the working of Generative Adversarial Network with proper architecture diagram [10]
- 4 a Write a short note on Bayesian Network with suitable example [10]
- b Draw and explain the architecture of Variational Autoencoder [10]
- 5 a Explain the concept of Metaverse [10]
- b Explain the two transfer learning approaches [10]
- 6 a Explain any three variants of Generative Adversarial Network [10]
- b Explain any three variants of Autoencoders [10]